

NARRABRI, Australia

WMO: 957340

Lat: 30.3153S

Lon: 149.8303E

Elev: 230

StdP: 98.60

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 0.5	(c) 2.0	(d) -7.7	(e) 2.0	(f) 17.4	(g) -5.5	(h) 2.4	(i) 15.9	(j) 10.7	(k) 17.2	(l) 9.5	(m) 17.3	(n) 1.1	(o) 170	(p) 0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 12.8	(c) 38.6	(d) 20.1	(e) 36.9	(f) 19.7	(g) 35.4	(h) 19.3	(i) 22.7	(j) 30.9	(k) 22.0	(l) 30.2	(m) 21.5	(n) 29.8	(o) 5.2	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	16.1	23.9	20.0	15.1	24.2	19.0	14.2	24.5	68.2	30.9	65.7	30.1	63.5	29.5	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.7	9.3	8.3	-2.3	41.3	1.2	2.6	-3.1	43.2	-3.8	44.7	-4.5	46.2	-5.3	48.0
			-3.1	24.2	1.2	1.2	-4.0	25.1	-4.7	25.8	-5.4	26.4	-6.3	27.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.4	27.6	26.4	23.7	19.7	15.0	12.0	11.1	12.2	15.8	20.2	23.3	26.2
	DBStd	6.60	3.39	3.25	2.85	2.91	2.83	2.59	2.68	2.99	3.37	3.61	3.71	3.41
	HDD10.0	42	0	0	0	0	2	11	18	10	1	0	0	0
	HDD18.3	850	0	0	1	18	108	189	224	192	91	22	4	1
	CDD10.0	3468	544	459	426	291	155	72	53	79	175	315	398	502
	CDD18.3	1235	286	226	169	59	4	0	0	2	15	79	152	244
	CDH23.3	14642	3577	2693	1728	553	49	0	1	21	247	934	1798	3039
	CDH26.7	7154	1979	1420	732	132	2	0	0	3	65	342	854	1625
Wind	WSAvg	4.0	4.5	4.3	4.0	3.7	3.1	3.4	3.3	3.6	4.1	4.4	4.6	4.8
Precipitation	PrecAvg	632	111	67	43	40	64	32	43	34	33	44	46	66
	PrecMax	901	296	220	137	175	208	100	157	126	105	141	103	166
	PrecMin	321	0	0	0	0	0	0	0	0	0	0	0	3
	PrecStd	189	88	59	40	49	58	25	37	33	32	37	33	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	40.4	36.2	31.9	26.5	22.1	22.8	25.7	31.5	34.7	39.7	40.3
		MCWB	21.1	20.8	19.7	17.3	15.5	13.6	12.9	13.9	15.3	16.9	19.0	19.1
	2%	DB	38.4	37.8	34.2	29.7	24.7	20.5	20.7	23.2	28.5	32.3	36.0	37.5
		MCWB	20.7	20.0	19.2	17.4	14.8	13.1	12.5	13.1	14.8	16.9	18.2	19.2
	5%	DB	36.6	35.9	32.4	28.1	23.3	19.2	19.0	21.4	26.1	30.3	33.6	35.9
		MCWB	20.2	19.8	18.8	16.7	14.1	12.6	12.0	12.2	14.1	16.3	17.9	19.0
	10%	DB	34.9	33.7	30.7	26.6	21.9	17.8	17.5	19.7	24.0	28.3	31.4	34.1
		MCWB	20.0	19.4	18.7	16.2	13.6	12.1	11.5	11.6	13.6	16.0	17.5	18.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.9	24.0	22.4	20.4	17.9	16.1	16.7	16.5	18.1	20.1	21.8	22.6
		MCDB	31.9	32.0	29.0	25.9	21.8	17.8	19.0	23.3	25.6	27.8	30.4	29.5
	2%	WB	22.8	22.7	21.6	19.0	16.5	14.8	14.4	14.8	16.9	18.8	20.6	21.7
		MCDB	30.7	31.7	28.1	26.0	21.8	17.8	18.2	19.1	23.4	27.1	29.2	29.6
	5%	WB	22.1	21.8	20.8	17.9	15.4	13.8	12.9	13.5	15.8	17.9	19.8	21.0
		MCDB	30.7	30.1	27.7	24.7	21.2	17.5	17.2	19.4	22.4	26.5	28.1	29.7
	10%	WB	21.5	21.0	20.0	17.1	14.3	12.9	11.9	12.4	14.7	17.0	19.1	20.4
		MCDB	30.4	29.8	27.5	24.5	20.5	16.9	16.8	18.6	22.2	25.5	27.6	29.2
Mean Daily Temperature Range	5% DB	MDBR	12.8	13.2	12.8	13.5	13.7	11.4	12.7	14.2	15.2	14.5	13.8	13.8
		MCDBR	14.3	15.4	14.6	14.7	14.7	13.0	14.5	16.4	17.9	16.8	16.6	15.4
		MCWBR	4.5	4.8	5.3	5.9	6.9	6.9	8.0	8.6	8.2	6.9	5.9	5.4
	5% WB	MCDBR	12.0	12.5	11.7	12.3	11.9	10.0	11.6	12.8	13.8	13.6	12.5	12.5
		MCWBR	4.2	4.4	4.7	5.3	5.9	5.9	7.0	7.3	7.4	6.1	4.9	4.9
Clear-Sky Solar Irradiance	taub		0.360	0.361	0.334	0.315	0.296	0.290	0.285	0.297	0.330	0.349	0.373	0.361
	taud		2.532	2.540	2.609	2.618	2.616	2.641	2.637	2.575	2.474	2.458	2.439	2.504
	Ebn at Noon		980	961	956	925	898	881	903	931	943	961	961	981
	Edh at Noon		111	107	94	84	75	69	73	86	106	115	121	115
All-Sky Solar Radiation	RadAvg		7.43	6.75	5.75	4.74	3.69	2.97	3.37	4.33	5.54	6.49	6.98	7.44
	RadStd		0.45	0.49	0.37	0.25	0.20	0.25	0.21	0.26	0.40	0.50	0.69	0.55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+227	+150

Nomenclature: See separate page